

## Laws of Chemical Change

1. Magnesium oxide is formed in three separate experiments by the reaction of magnesium and oxygen.

|            | Experiment #1 | Experiment #2 | Experiment #3 |
|------------|---------------|---------------|---------------|
| Mass of Mg | 0.0628 g      | 0.0411 g      | 0.0830 g      |
| Mass of O  | 0.0415 g      | 0.0270 g      | 0.0550 g      |

*Calculate the ratio of Mg:O for each experiment and state which law is illustrated by these results.*

2. Iron forms two different compounds with chlorine. These are 44.0 g of iron in 100.0 g one compound but only 34.4 g of iron in 100.0 g of the other compound. Find the Fe:Cl for each compound and state which law is demonstrated.
3. Nickel forms several oxides. One contains 71.0% nickel and another 78.6% nickel. Find the Ni:O and state the law.
4. Show by calculation that the following statements are equivalent and state the law.
- Iron combines with oxygen in the same mass ratio, 3.49 : 1.00
  - A compound of iron and oxygen contains 77.8% iron by mass.
  - A mass of 0.186 g of iron and 0.0533 g of oxygen combine to form a compound.
5. Copper and sulfur react together in a manner similar to zinc and sulfur. A known mass of copper powder is mixed with excess sulfur and the mixture is heated to induce a reaction. Afterward the unreacted sulfur is allowed to burn away to SO<sub>2</sub> which escapes.

From the following data, calculate the mass ratio of Cu:S

|                         |          |
|-------------------------|----------|
| Mass of test tube       | 21.511 g |
| Mass of test tube & Cu  | 22.524 g |
| Mass of test tube & CuS | 22.776 g |

6. Show that the following sets of data are in accord with the Law of Definite Proportions:
- 6.54 g Zn + 3.21 g S → 9.75 g ZnS
  - 7.00 g Zn + 3.21 g S → 9.75 g ZnS + 0.46 g Zn
  - 6.54 g Zn + 4.00 g S → 9.75 g ZnS + 0.79 g S

*What other law is illustrated here?*

7. Given that 3.04 g of pure zinc sulphide (ZnS) contains 1.00 g S and the remainder Zn, calculate the mass of S that would combine with:>
- 1.0 g Zn
  - 40.0 g Zn